DAY 29 | POST-READ

Evaluating GenAI Systems: Recap and Beyond

What we covered today, plus ideas to stretch your understanding.

Prepared by **Ahmed** | linkedin.com/in/ahmed-al-imaad

Part A: Summary of today

Topic	What to remember
Evaluation loop	Dataset, run, score, analyse failures, improve one thing, repeat.
Eval set	Realistic questions with expected answers, sources, and tags (topic, language).
Retrieval metrics	Hit rate at k, MRR, context relevance.
Generation metrics	Faithfulness (supported claims / all claims), answer relevance, correctness.
Error analysis	Retrieval miss vs generation error vs correct.
Hallucination types	Unsupported addition, contradiction, wrong source mixing, fabricated citation.
LLM as judge	Rubric, one criterion per call, reasons first, calibrate with humans.
Judge biases	Position, verbosity, self-preference, leniency.
Bias checks	Counterfactual swaps, slice metrics, representation of groups in the eval set.

Part B: Going beyond today

1. Ragas

```
from ragas import evaluate
from ragas.metrics import faithfulness, answer_relevancy, context_precision, context_recall
```

Ragas packages common RAG metrics using LLM judges. Use it once you understand what each metric measures, as you now do. Check its documentation for the current API.

2. Synthetic test generation

Models can draft questions from your documents to grow an eval set quickly. Always have people review them; synthetic sets tend to be easier and more literal than real user questions.

3. Inter-annotator agreement

Before trusting human labels, have two people label the same 30 items and compare. If humans disagree often, the rubric is unclear, and no judge model will fix that.

4. Online evaluation

Offline sets catch regressions before release. After release, sample real conversations, collect thumbs up and down, and add confusing real questions to the eval set.

5. Multilingual quality

Embedding models and LLMs vary widely by language. If users write in Japanese or Hindi, include those languages in the eval set and choose models that handle them well.

Part C: Check yourself

- Why evaluate retrieval and generation separately?
- How is faithfulness calculated?
- Name two LLM judge biases and how to reduce them.
- What does slicing results by language reveal?
- What is the difference between a retrieval miss and a generation error?

Part D: Practice ideas for tonight

- Add 10 real-style questions, including 3 in Japanese or Hindi, to the eval set.
- Label 15 answers yourself and compare with the offline or LLM judge.
- Write a short evaluation report: overall scores, slices, top three failure causes and fixes.

Coming up on Day 30: golden datasets, prompt regression testing, context precision and recall, BLEU, and quality gates in CI.

Otsukaresama deshita (お疲れ様でした): thank you for your hard work today